Physical Chemistry
February 10, 2006

180 grams of H_2O steam (M_0) is suddenly cooled down from 125°C (T_1) to 20°C (T_2), and then be immediately sealed inside a container. The container does not allow any work or heat exchange between the over-cooled steam and the surroundings.

The heat capacity, $C_{v,water}$, of water is 75 (J K^{-1} mol^{-1}) and the heat capacity of the steam, $C_{v,steam}$, is 35 (J K^{-1} mol^{-1}). Both heat capacity values are not temperature dependent. The boiling point of water is 100°C (T_3) under current conditions. ΔU_b of the vaporization is 40 (kJ mol^{-1}) and also not temperature dependent.

(1) Qualitatively predict what will happen to the steam after the processes described above, using the first law and the second law of thermodynamics. Predict the maximum value of the final equilibrium temperature (T_3) of the system. If you cannot do this step after thinking for more than 10 minutes, go to the second step.
(2) Using the first law of thermodynamics to calculate the final temperature (T_3) of the system, assuming all the steam will suddenly condense to liquid.
(3) Do you think the final temperature calculated in step (2) is reasonable? Why? Can you answer finish the first step now? If you cannot, very likely, you will not pass this exam. Anyhow, you may still try it. It never hurts.
(4) Calculate the molar fraction of the steam (X) will condense, based on the final equilibrium temperature (T_3) determined in the third step. Again, assuming that all the necessary condensation occurs instantly and the system warms up by the energy released in the condensation process.
(5) For an isolated system, the second law of thermodynamics requires that the randomness of the system during a spontaneous process should increase. However, your results indicate that a system can spontaneously generate liquid from gas, and thus decrease its structural randomness. Do you think your results contradict with the second law of thermodynamics? If yes, what is your explanation? If not, please qualitatively identify whatever entropy change which may compensate this structural entropy decrease.